

INFORMATION

Portfolio

LinkedIn

villanimatteo33@gmail.com

Barcelona, Spain

TECHNICAL SKILLS

- **Robotics**: ROS 2, BehaviorTree, motion planning (RoboPlan, MoveIt2), grasp pose detection, arm/camera calibration, 3D reconstruction.
- **AI/ML:** VLA — policy train & deploy, RL— reward shaping, Deep Learning, computer vision, LSTM.
- **Simulation**: Gazebo, MuJoCo, Rviz
- **Software:** C++, Python, Docker, CI/CD, Pixi, Git, Linux.

LANGUAGES

- Italian — Native
- English — B2
- Spanish — B2

MATTEO VILLANI

ROBOTIC SOFTWARE ENGINEER

ROS2 · MANIPULATION · ROBOT LEARNING

PROFILE

Robotics software engineer at PAL Robotics building ROS 2 manipulation systems for real robots: language-prompted perception, grasp planning and behavior-tree orchestration, validated in simulation and on hardware across TIAGo, TIAGo++ and TIAGo Pro. Worked on the path from recorded robot data to policies running on a Jetson. MSc in AI & Robotics (110/110). Looking to take robust AI-driven manipulation into production Robots.

PROFESSIONAL EXPERIENCE

- **ROBOTIC SOFTWARE ENGINEER - PAL ROBOTICS**
  Barcelona 2024 - Present
  - Designed and built **GAM**, a modular ROS 2 grasping pipeline: perception (GroudingDINO + SAM), 3D object reconstruction, grasp-pose detection (GPD) and staged motion planning (MTC), orchestrated with behavior trees — delivering end-to-end autonomous manipulation on the TIAGo Family.
  - Built single-arm, bimanual and multi-robot coordination demos & PoC, by designing the system architecture and writing behavior-tree nodes that implement desired behaviors, plus REST APIs for robot communication.
  - Ran the full VLA pipeline from teleop demonstrations to policy deployment on the TiagoPro's Jetson Orin, using Pi0.
  - Drove end-to-end integration & validation of external AI frameworks (eg., MCC, Yolo..) in simulation and real hardware; iterated to enhance the TIAGo robots' capabilities (eg., perception, detection..).
  - Integrated and evaluated third-party components with external partners in EU-funded projects.

- **ML SOFTWARE DEVELOPER- RIATLAS**
  Salerno, 2021-2022
  - Developed a medical visualization application to improve specialists' monitoring of patient therapy progress. Solved a *Human-activity recognition* problem designing and implementing an LSTM-based ML model in a 3+ member team.

EDUCATION

- **MASTER OF COMPUTER ENGINEERING** 2022 - 2025
  Artificial Intelligence & Robotics 110/110
  University of Padua, IT - *graduated in July 2025*
  - ERASMUS PROGRAM PARTICIPATION *(Feb 2024 - July 2024)*
    Artificial Intelligence & Advanced Robotics
    Graz University of Technology, AT

- **BACHELOR OF COMPUTER ENGINEERING** 2019 - 2022
  Computer Engineering
  University of Salerno, IT - *graduated in Sept 2022*